



blocks of LIFE

are made. Out of these, 25 are essential for life, and six are used as the building blocks for all living things—carbon, hydrogen, oxygen, nitrogen, sulphur, and phosphorus.

INDISPENSABLE DNA

DNA is found in every cell in our bodies, and carries the information that is needed to make each cell function and reproduce properly. DNA is like an instruction manual that's all in code. It is made up of four chemicals, called nucleotides. These are adenine, cytosine, thymine, and guanine. When making the DNA code, adenine always pairs up with thymine and cytosine pairs with guanine. They are arranged in pairs in spiral chains that look like a twisted ladder, which is known as a double helix.

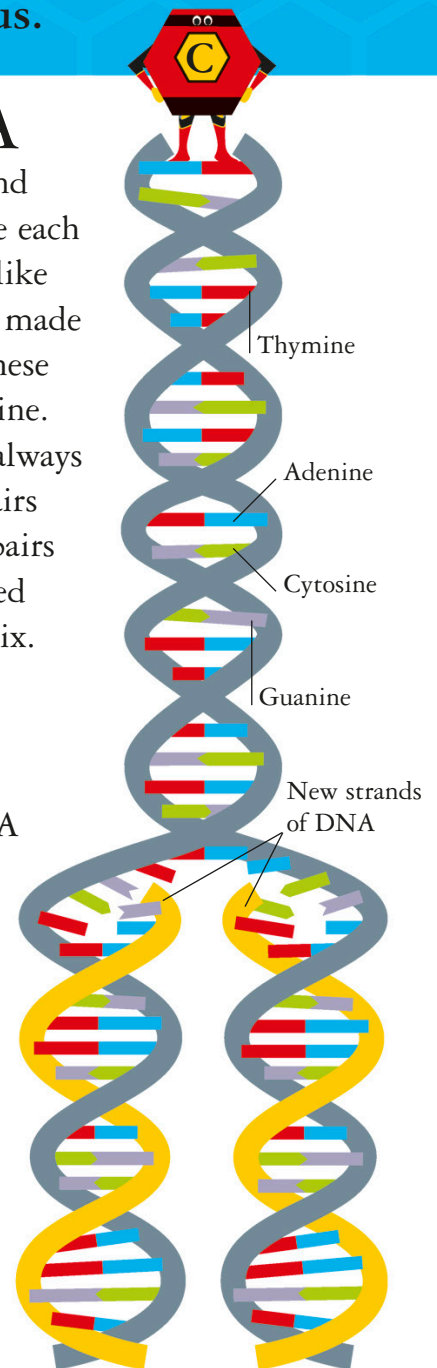
REPRODUCING ITSELF

Every time a cell needs to reproduce, the DNA unzips along the middle of its “rungs” into two strands. Each of these strands then creates the other side of the “ladder.”

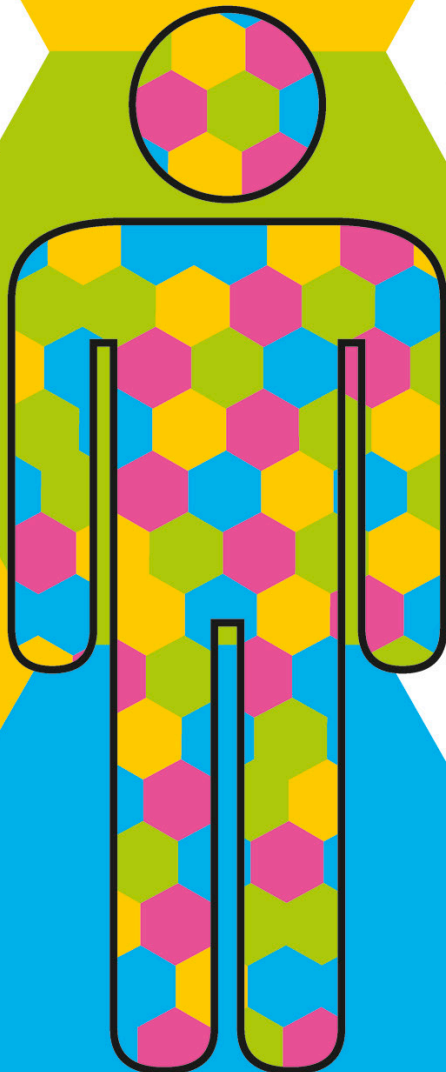
When the process is finished there is an exact copy of the original DNA chain.

MAKING PROTEINS

DNA is also used to make new proteins for a cell. When a new protein is required, the section of DNA that is the code for that protein unzips and is copied by a molecule called messenger RNA. The messenger RNA moves the copied section to another part of the cell to make the protein.



When DNA splits, each strand acts as a template to make the other side of the helix because the nucleotides always pair up with their partner.



The human body is made up of hundreds of different carbon compounds.